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Techno-Economic and Environmental Assessment of Rooftop Photovoltaic Systems in Baghdad Using SketchUp-PVsyst Simulation

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Abstract: Rooftop photovoltaic (PV) systems are a practical option for Baghdad, but their performance depends on module technology, mounting strategy, row spacing, and near-shading. This study integrates SketchUp/Skelion and PVsyst to evaluate grid-connected rooftop PV systems on the College of Education building at Mustansiriyah University, Baghdad. Seventeen configurations were simulated using monocrystalline and heterojunction (HJT) modules, fixed, seasonal, and two-axis mounting strategies, portrait and landscape layouts, row spacings of 0, 0.5, and 1.0 m, and practical losses including soiling, aging, unavailability, and auxiliary consumption. The objective was to combine a realistic 3D rooftop model with PVsyst near-shading analysis and compare technical, economic, and environmental performance. HJT modules outperformed comparable monocrystalline cases under Baghdad's hot conditions because of their lower power temperature coefficient. The two-axis HJT case achieved the highest annual energy production (272,308 kWh/yr), while the seasonal HJT case achieved the highest performance ratio (85.22%). The fixed landscape HJT system provided a strong practical balance, producing 244,305 kWh/yr and 1,871 kWh/kWp/yr with a performance ratio of 85.19%. Under the stated 25-year economic assumptions, its discounted levelized cost of energy (LCOE) was 0.0291 USD/kWh, with a simple payback period (PBP) of 3.1 years. Increasing row spacing improves specific yield by reducing self-shading, but total rooftop output decreases because fewer rows fit on the roof. Over the 25-year study period, the simulated systems avoid approximately 1,840.6–4,043.8 tCO_2 relative to grid electricity. Overall, the fixed landscape HJT arrangement is a practical candidate for rooftop PV deployment in Baghdad and similar hot, dry urban environments.

Keywords: Rooftop photovoltaic system; PVsyst; SketchUp; Heterojunction modules; Shading analysis; Techno-economic assessment; CO_2 reduction

1 Introduction

Iraq has substantial solar resources and a strong rationale for distributed photovoltaic (PV) deployment [1, 2]. Baghdad also experiences high ambient temperatures that reduce PV module output [3], while dust deposition can impose additional optical losses [4]. In dense urban rooftops, parapets, service rooms, neighboring structures, and row-to-row geometry can create near-shading that is difficult to represent with simple two-dimensional sizing methods [5, 6]. These factors make realistic rooftop geometry, thermal behavior, soiling, and shading important design considerations rather than secondary corrections.

Previous work has addressed complementary parts of this problem. Al-Ghezi et al. [2, 3] quantified Baghdad solar-resource and temperature effects, while Musa and Shneishil [6] used 3D shading and PVsyst for rooftop layout optimization in Baghdad. Ebhota and Tabakov [7], Moustafa et al. [8], and Kumar et al. [9] demonstrated the value of simulation tools for orientation, operating-temperature, and system-design studies. However, most studies examine a limited set of design variables or emphasize technical output without a consistent economic and environmental comparison at roof scale.

Module technology creates an additional design dimension. Zhang et al. [10] and Ali et al. [11] summarize the device-level and thermal-performance potential of advanced heterojunction (HJT) architectures, while Camilo and Santos [12] demonstrate the importance of reporting technical and financial indicators together. A gap therefore

remains for Baghdad rooftops: few studies jointly assess realistic 3D near-shading, module technology, mounting strategy, layout density, and their combined techno-economic and environmental consequences.

Accordingly, the present study addresses the following design question: which rooftop PV configuration gives the most practical balance between energy yield, financial performance, and realistic constructability for an urban building in Baghdad? To answer this question, a 3D rooftop model of the College of Education building at Mustansiriyah University was developed in SketchUp/Skelion and imported into PVsyst for near-shading analysis. Seventeen variants were then compared by combining monocrystalline and HJT modules, fixed/seasonal/two-axis mounting strategies, portrait/landscape layouts, row-spacing alternatives, and several loss combinations.

The main contribution of the paper is the integration of three normally separate perspectives in one framework: (i) realistic 3D rooftop and near-shading modelling for a Baghdad case study; (ii) side-by-side technical comparison of module technology, mounting strategy, orientation, and spacing; and (iii) translation of these differences into economic and environmental indicators. This framework supports a practical design recommendation for future rooftop PV deployment in Baghdad and comparable hot urban environments.

2 Materials and Methods

2.1 Study Site and Integrated Simulation Workflow

The study site is the College of Education building at Mustansiriyah University in Baghdad, Iraq (latitude 33.3° N, longitude 44.4° E). The available rooftop consists of two similar blocks with a combined gross roof area of about 968 m^2. The PVsyst simulations used Meteonorm 8.0 meteorological data for Baghdad. The annual global horizontal irradiation (GHI) used in the model is about 1,986.4 kWh/m^2/year, equivalent to an annual daily average of approximately 5.44 kWh/m^2/day. This high solar resource is consistent with local Baghdad studies [2]. At the same time, hot summer conditions and dust exposure make the site suitable for examining the trade-off between high irradiance, thermal losses, soiling, and practical rooftop limitations.

The workflow began with the construction of a detailed roof model in SketchUp. The digital model included the principal rooftop obstructions and nearby geometries needed to reproduce the installation conditions. The Skelion plug-in was used to place PV rows and configure their spacing and tilt, after which the scene was exported to PVsyst. The same geometry was then imported into the PVsyst Near Shadings environment so that geometric and beam-shading effects could be included consistently in the annual simulations.

2.2 Selected Modules, Inverter, and Configuration Matrix

From the PVsyst database, two 680 W module families were selected: an n-type monocrystalline module (RCM-680-8DBNM) and an HJT module (RCM-680-8DBHM), whose simplified schematic is shown in Figure 1. Both modules have the same rated peak power, making the comparison more meaningful because differences in annual output can be attributed primarily to thermal behavior and system configuration rather than nominal module power. The selected inverter family was an SMA 30 kW string inverter, scaled appropriately for the varying array capacities. The main characteristics of the selected modules are compared in Table 1. The complete simulation workflow is summarized in Figure 2, while Figures 3–5 illustrate the rooftop geometry and corresponding simulation stages.

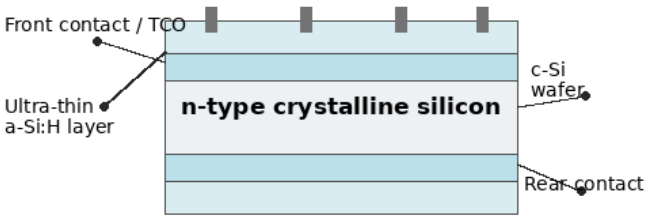


Figure 1. Simplified original schematic of a heterojunction (HJT) solar cell prepared for this study

The configurations were arranged so that the impact of module technology, mounting strategy, orientation, row spacing, and cumulative losses could be isolated. For the layout-only variants, the landscape cases accommodate 192 modules at 0 m spacing, 154 modules at 0.5 m spacing, and 116 modules at 1.0 m spacing, whereas the portrait cases accommodate 162, 147, and 120 modules, respectively. This difference in installed direct current (DC) capacity is important because wider row spacing reduces self-shading but also decreases the number of modules that can fit on the rooftop.

Table 1. Main electrical characteristics of the selected 680 W mono-Si and heterojunction (HJT) modules

Parameter	Mono-Si (RCM-680-8DBNM)	HJT (RCM-680-8DBHM)
Cell type	N-type monocrystalline	HJT
Maximum power, Pmax	680 W	680 W
Voltage at Pmax	39.7 V	42.08 V
Current at Pmax	17.13 A	16.16 A
Open-circuit voltage, Voc	47.5 V	49.2 V
Short-circuit current, Isc	18.11 A	17.18 A
Pmax temperature coefficient	-0.30%/°C	-0.24%/°C
Voc temperature coefficient	-0.25%/°C	-0.22%/°C
Isc temperature coefficient	+0.046%/°C	+0.047%/°C
Operating temperature	-40 to +85 °C	-40 to +85 °C
Nominal module operating temperature (NMOT)	43 ± 2 °C	42 ± 2 °C
Dimensions	2384 × 1303 × 35 mm	2384 × 1303 × 35 mm

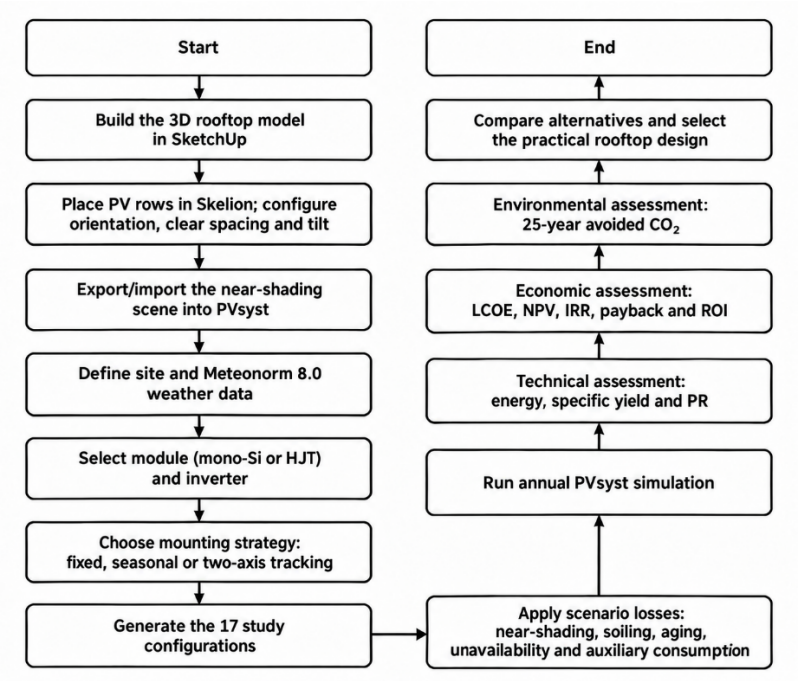


Figure 2. Workflow of the integrated SketchUp/Skelion-PVsyst methodology

Note: HJT = heterojunction technology; LCOE = levelized cost of energy; NPV = net present value; IRR = internal rate of return; ROI = return on investment; PR = performance ratio.

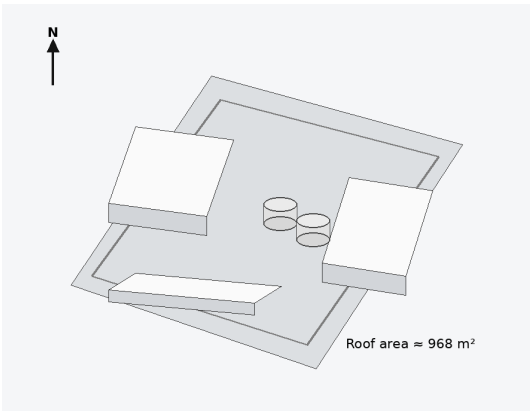


Figure 3. Author-generated schematic of the case-study rooftop geometry modeled in SketchUp

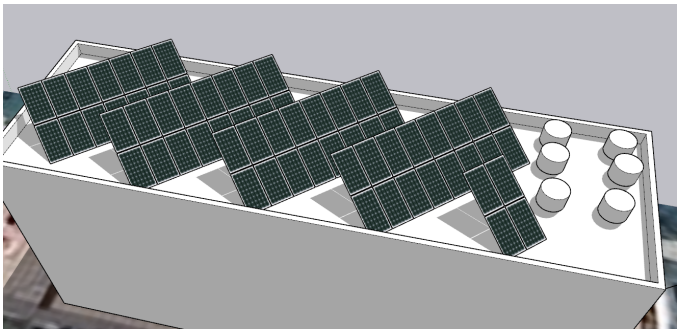


Figure 4. Example of a photovoltaic (PV) layout generated on the rooftop model

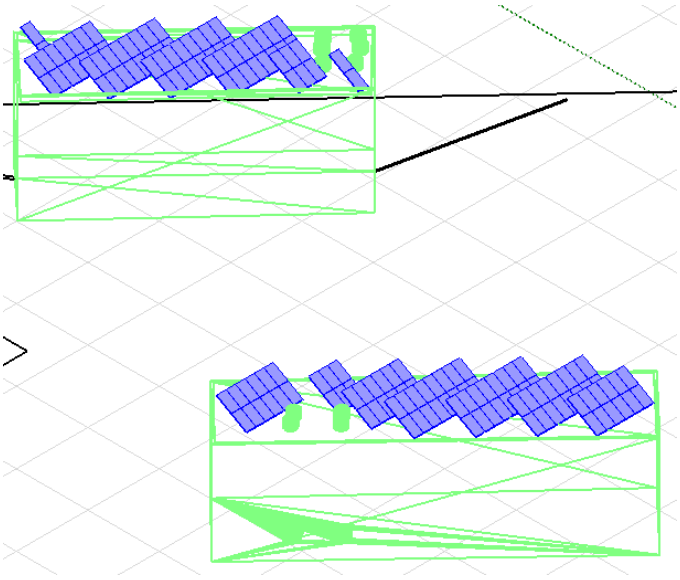


Figure 5. Imported near-shading scene in PVsyst

2.3 Row-Spacing Criterion and Variant Naming

The row-spacing check was based on simple shadow geometry for a tilted PV row [13]. Let L be the sloped module dimension in the north-south row section, β the module tilt, and $\alpha_{\min}$ the critical solar elevation angle. The minimum clear horizontal distance required for the shadow from the upper edge of one row to reach the lower edge of the next row, together with the corresponding minimum horizontal pitch, are given by:

$$D_{\min} = \frac{L\sin(\beta)}{\tan(\alpha_{\min})}, \qquad P_{\mathrm{row,min}} = L\cos(\beta) + D_{\min} \tag{1}$$

Here, $D_{\min}$ is the shadow-avoidance clear distance between adjacent rows and $P_{\mathrm{row,min}}$ is the horizontal pitch between corresponding lower edges of successive rows. For the fixed landscape-layout geometric check, L = 1.303 m, β = 30°, and $\alpha_{\min} \approx 33.3°$ (approximately the solar-noon elevation at the winter solstice for latitude 33.3° N). These values give $D_{\min} \approx 0.99$ m and $P_{\mathrm{row,min}} \approx 2.12$ m. The simulated clear-spacing scenarios of 0, 0.5, and 1.0 m were implemented directly as design alternatives in the rooftop layout. Thus, the 0 and 0.5 m cases intentionally accept greater self-shading in exchange for higher installed capacity, whereas the 1.0 m case is close to the geometric shadow-avoidance distance at the selected critical condition. The definitions of all 17 simulated PV system variants are summarized in Table 2. The seasonal HJT case is denoted as HLSN consistently throughout the text and figures.

2.4 Loss Model and Simulation Assumptions

To move the simulations away from idealized output and closer to practical operation, several additional losses were introduced stepwise. The base soiling loss was set to 3%. Module aging was represented using the PVsyst degradation setting corresponding to year 10 of operation. System unavailability was set to 2%, equivalent to approximately 7.3 days/year, and auxiliary electricity consumption was included in the fully loaded cases. These terms were accumulated progressively in the loss-build-up variants so that their combined impact could be observed.

Standard PVsyst thermal, electrical, mismatch, wiring, and inverter losses remained active according to the selected components and system model. The geometric definition of inter-row shading variables used in this analysis is illustrated in Figure 6.

Table 2. Definition of the 17 photovoltaic (PV) system variants

No.	Full Configuration	Code Used in Text
1	Mono, landscape, fixed tilt; dense 0 m layout; controlled comparison	MLF
2	Heterojunction (HJT), landscape, fixed tilt; dense 0 m layout; controlled comparison	HLF
3	HJT, landscape, seasonal tilt; dense 0 m layout; controlled comparison	HLSN
4	HJT, landscape, two-axis tracking; same 192-module basis; controlled comparison	HLT
5	HJT, portrait, 0 m spacing; near-shading only	HPS
6	HJT, portrait, 0.5 m spacing; near-shading only	HP05S
7	HJT, portrait, 1.0 m spacing; near-shading only	HP1S
8	HJT, landscape, 0 m spacing; near-shading only	HLS
9	HJT, landscape, 0.5 m spacing; near-shading only	HL05S
10	HJT, landscape, 1.0 m spacing; near-shading only	HL1S
11	HJT, landscape, 0 m; shading + soiling	HLSS
12	HJT, landscape, 0 m; shading + soiling + aging	HLSSA
13	HJT, landscape, 0 m; shading + soiling + aging + unavailability	HLSSAU
14	HJT, landscape, 0 m; shading + soiling + aging + unavailability + auxiliaries	HLSSAUA
15	HJT, landscape, 0.5 m + all added losses	HL05SSAUA
16	HJT, landscape, 1.0 m + all added losses	HL1SSAUA
17	Mono, landscape, 0 m + all added losses	MLSSAUA

Notes: "Controlled comparison" means that the extra scenario-specific near-shading, soiling, aging, unavailability, and auxiliary terms used in the later loss-build-up cases are not applied; standard PVsyst thermal, electrical, wiring, mismatch, and inverter losses remain active. The first three controlled cases use the dense 0 m clear-spacing layout. HLT retains the same 192-module capacity basis for the mounting-strategy comparison. HPS–HL1S include the near-shading scenario only; HLSS–HLSSAUA use 0 m spacing and progressively add the stated losses; HL05SSAUA and HL1SSAUA use 0.5 and 1.0 m spacing with all added losses.

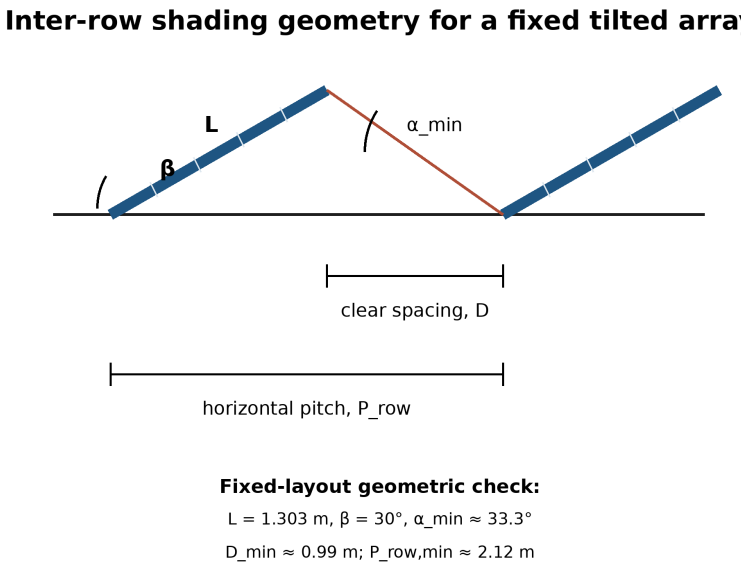


Figure 6. Geometric interpretation of clear row spacing and horizontal pitch used in the layout analysis

Figure 7 and Figure 8 show two of the mounting options implemented in PVsyst: seasonal tilt and two-axis tracking.

The first four variants keep the installed DC capacity constant at 192 modules (130.56 kWp) so that module technology and mounting strategy can be compared without the confounding effect of array size. They use the same nominal dense-layout basis, and the additional near-shading/soiling/aging/unavailability/auxiliary scenario terms used in the later loss-build-up cases are not applied to this controlled comparison.

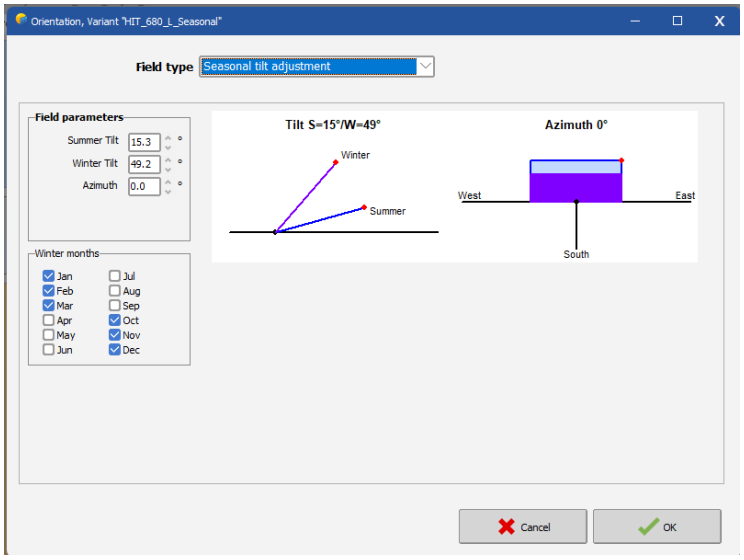


Figure 7. Example of the seasonal-tilt definition used in PVsyst

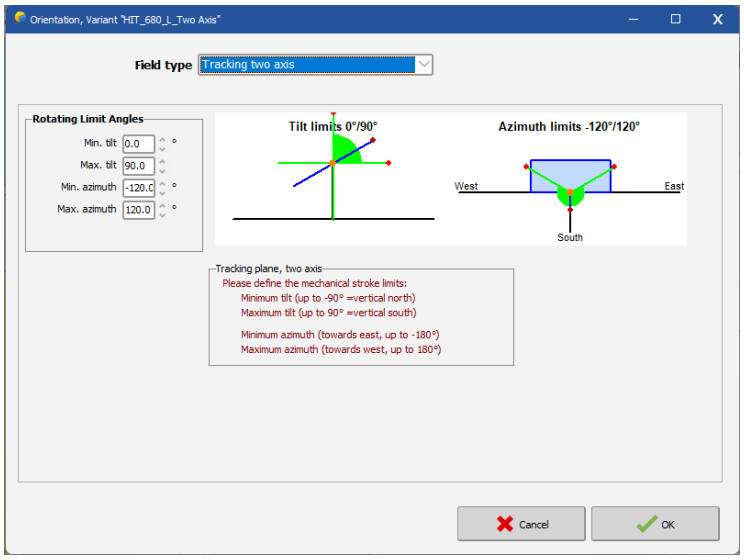


Figure 8. Example of the two-axis tracking definition used in PVsyst

2.5 Technical Performance Indicators

The technical assessment follows standard PV performance definitions commonly used in PV performance studies [14]. Four indicators were used: annual alternating current (AC) energy production, reference yield, final yield, and performance ratio.

$$E_{\mathrm{AC}} = \sum_{t=1}^{N} P_{\mathrm{AC},t} \Delta t \tag{2}$$

$$Y_r = \frac{H_{\mathrm{POA}}}{G_{\mathrm{STC}}} \tag{3}$$

$$Y_f = \frac{E_{\mathrm{AC}}}{P_0} \tag{4}$$

$$PR = \left(\frac{Y_f}{Y_r} \right) \times 100\% \tag{5}$$

where, E_{AC} is the annual grid-connected AC electrical energy (kWh), $P_{\mathrm{AC},t}$ is the AC output power (kW) at annual simulation time step t, Δt is the time-step duration in hours, and N is the total number of time steps in one year. H_{POA} is the annual plane-of-array irradiation (kWh/m^2), G_{STC} is the standard-test-condition irradiance (1 kW/m^2), and P_0 is the installed DC capacity (kWp). Y_r represents the available solar resource, Y_f is the useful annual AC energy per unit of installed capacity, and PR indicates how effectively the system converts the available plane-of-array irradiation into useful AC energy after modeled losses.

2.6 Economic and Environmental Assessment

The financial evaluation used net present value (NPV), internal rate of return (IRR), simple payback period (PBP), a discounted NPV-based Return on Investment (ROI), and Levelized Cost of Energy (LCOE) [12]. Each configuration used its reported initial investment and the common base-case assumptions in Table 3. Public vendor quotations for the selected module and inverter families were used only as price benchmarks; the comparative calculations use the total investment values reported for each variant.

Table 3. Base-case economic and environmental assumptions used for the comparative analysis

Parameter	Base-Case Value Used in the Analysis
Analysis period	25 years
Discount rate, a	2%
Effective displaced electricity value, V_e	0.063 USD/kWh
Annual O&M cost	1% of initial investment
Salvage value at year n	0
Currency basis	2025 USD
Grid emission factor, EF_{grid}	0.594 kgCO_2/kWh (adopted base-case value)
Electricity-value escalation	0%/yr
Cash-flow energy treatment	Constant annual PVsyst result for each variant; no additional annual degradation beyond the selected PVsyst aging case

Note: O&M = operation and maintenance.

The initial investment C_0 includes PV modules, inverter units, mounting structure, DC and AC wiring, protection and connection components, installation and commissioning, and other balance-of-system items connecting the rooftop system to the grid. For the two-axis tracking case, the tracking structure, drive/control system, and related installation cost were also included. Land purchase, taxes, financing charges, battery storage, roof strengthening, and major component replacement were outside the cost boundary. The same boundary was applied consistently across configurations.

$$\mathrm{NPV} = -C_0 + \sum_{j=1}^{n} \frac{C_j}{(1+a)^j} \tag{6}$$

$$\mathrm{LCOE} = \frac{C_0 + \sum_{j=1}^{n} \frac{C_{\mathrm{O\&M},j}}{(1+a)^j}}{\sum_{j=1}^{n} \frac{E_j}{(1+a)^j}} \tag{7}$$

$$\mathrm{PBP} = \frac{C_0}{C_{\mathrm{net,annual}}}; \qquad \mathrm{ROI} = \frac{\mathrm{NPV}}{C_0} \times 100\%; \qquad \mathrm{IRR}: \ \mathrm{NPV}(r) = 0 \tag{8}$$

Here, C_0 is the initial investment, a is the discount rate (0.02 in the base case), n is the 25-year analysis period, $C_{\mathrm{O\&M},j}$ is the operation and maintenance (O&M) cost in year j, E_j is the annual energy used in the cash-flow model, and C_j is the annual net cash flow. Under the stated base case, $C_j = E_j V_e - C_{\mathrm{O\&M},j}$, where $V_e = 0.063$ USD/kWh and $C_{\mathrm{O\&M},j} = 0.01\, C_0$ per year. The simple PBP is undiscounted and uses the constant annual net cash flow. The ROI reported here is explicitly $NPV/C_0 \times 100\%$, so it is a discounted NPV-based ratio rather than an undiscounted lifetime ROI. IRR is the discount rate r that makes Eq. (6) equal to zero. No electricity-price escalation or additional year-by-year degradation was applied in the financial model; where the PVsyst aging setting is included, its effect is already embedded in the corresponding annual-energy result.

$$\mathrm{CO_{2,avoided}}\ (\mathrm{tCO_2}) = \frac{1}{1000} \sum_{j=1}^{n} (E_j \times EF_{\mathrm{grid}}) \tag{9}$$

Eq. (9) was used to calculate the avoided CO_2 values reported in Table 5. EF_{grid} is the adopted base-case grid emission factor of 0.594 kgCO_2/kWh, and the factor 1/1000 converts kilograms to tonnes. The lifetime calculation uses the annual energy value for each variant over the 25-year study period. No additional year-by-year degradation correction is added to the CO_2 summation; cases that include the PVsyst aging-loss setting already contain that penalty in their annual-energy result.

3 Results and Discussion

3.1 Reference Shading and Loss Analysis

Before comparing all variants, two complementary simulation outputs were used to interpret the results. Figure 9 shows the sun-path and shading analysis for the fully loaded HJT landscape case with 1.0 m spacing (HL1SSAUA), illustrating annual solar access for the modeled roof. Figure 10 shows the PVsyst loss diagram for the dense 0 m all-loss HJT landscape case (HLSSAUA); its grid injection of 181,982 kWh matches Table 4. The sun-path plot indicates that many surrounding obstructions act mainly at low solar angles, while the annual loss diagram quantifies the combined near-shading effect for the modeled geometry.

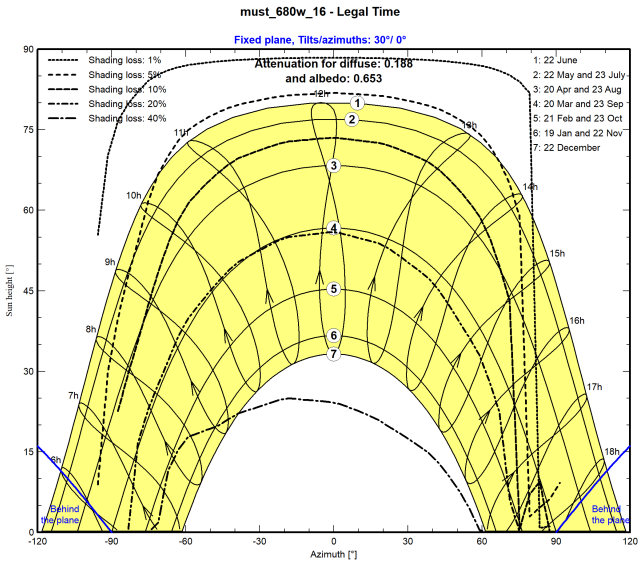


Figure 9. Sun-path and shading analysis for the HL1SSAUA reference configuration

Table 4. Technical performance of the 17 photovoltaic (PV) variants

Variant	Installed DC Capacity (kWp)	Produced Energy (kWh/yr)	Specific Production (kWh/kWp/yr)	PR (%)
MLF	130.56	239,314	1,833	83.44
HLF	130.56	244,305	1,871	85.19
HLSN	130.56	253,304	1,940	85.22
HLT	130.56	272,308	2,086	73.63
HPS	110.16	181,510	1,648	75.01
HP05S	99.96	171,465	1,715	78.09
HP1S	81.60	144,888	1,776	80.83
HLS	130.56	201,576	1,544	70.29
HL05S	104.72	177,922	1,699	77.35
HL1S	78.88	138,335	1,754	79.84
HLSS	130.56	195,810	1,500	68.28
HLSSA	130.56	185,522	1,421	64.69
HLSSAU	130.56	182,410	1,397	63.60
HLSSAUA	130.56	181,982	1,394	63.45
HL05SSAUA	104.72	160,643	1,534	69.83
HL1SSAUA	78.88	123,944	1,571	71.53
MLSSAUA	130.56	179,586	1,376	62.62

Note: DC = direct current; PR = performance ratio.

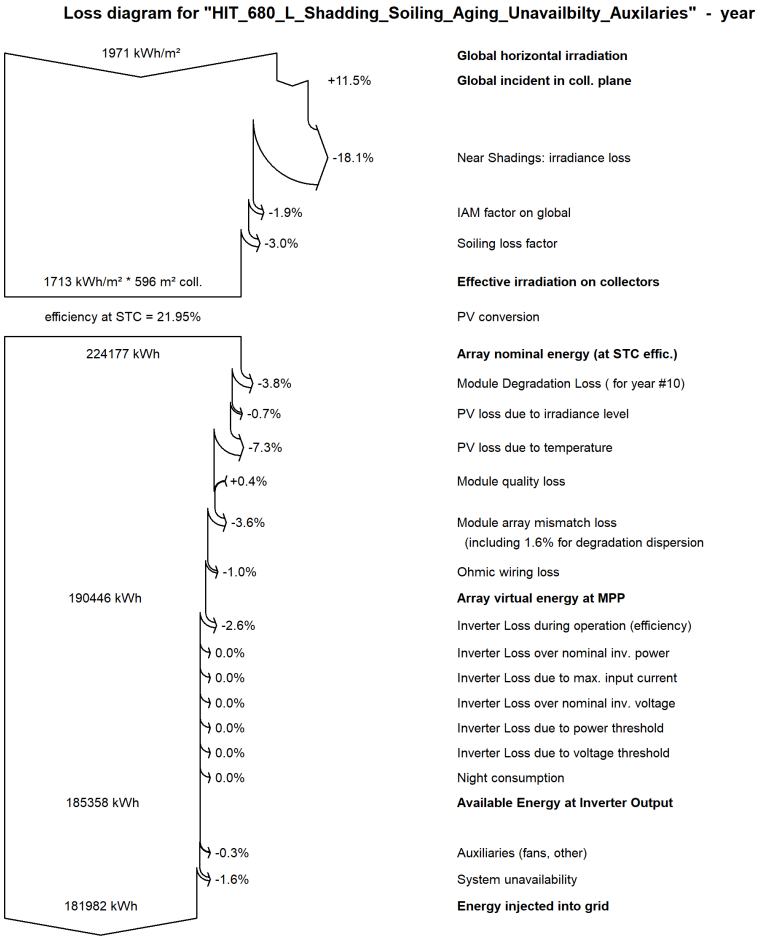


Figure 10. Loss diagram for the HLSSAUA dense 0 m all-loss reference configuration

Figure 10 shows that near-shading irradiance loss is approximately 18.1%, larger than the approximately 7.3% temperature loss in this dense all-loss case. The near-shading term is an aggregate geometric loss from row-to-row self-shading together with the modeled rooftop and surrounding obstructions; the exported loss diagram does not separate these contributions numerically. Additional penalties include the 3% soiling assumption, electrical and inverter losses, aging, auxiliaries, and unavailability. Consequently, the results do not support treating surrounding-building shading alone as the dominant loss. Instead, dense row geometry and high operating temperature are both important in Baghdad, and the relative importance changes with layout spacing and loss scenario.

3.2 Technical Comparison of the 17 Variants

Table 4 summarizes the technical results of the 17 simulated configurations. Installed DC capacity is calculated directly from the module count using 0.680 kWp per module, and the specific production values are reported relative to those exact capacities. Figures 11–14 visualize the monthly behavior of the principal groups of variants.

The two-axis tracking HJT case (HLT) produced the highest annual energy output, 272,308 kWh/yr, corresponding to 2,086 kWh/kWp/yr. Tracking increases plane-of-array irradiation throughout the day and year. Its performance ratio (73.63%) is nevertheless lower than those of the fixed and seasonal HJT cases because the reference yield also increases while thermal and system losses remain. The HLSN gives the highest PR (85.22%), while the fixed landscape HJT case (HLF) is very close at 85.19%. This is practically important because a simple fixed structure can provide nearly the same conversion-quality indicator without requiring seasonal mechanical adjustment.

The HJT advantage over the selected monocrystalline module is also visible in the controlled comparison. At the same installed capacity of 130.56 kWp, annual energy increases from 239,314 kWh/yr for MLF to 244,305 kWh/yr for HLF, while PR rises from 83.44% to 85.19%. This difference is consistent with the lower Pmax temperature coefficient of the HJT module in Table 1 (-0.24%/°C versus -0.30%/°C). Under Baghdad's high summer temperatures, the lower temperature sensitivity helps the HJT module retain higher specific production and PR.

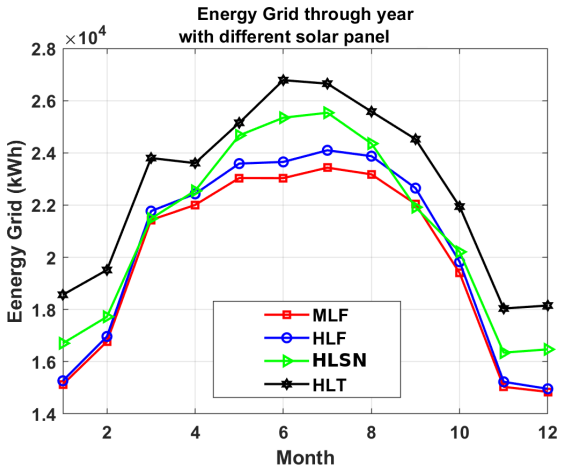


Figure 11. Monthly grid energy for the module-type and mounting-strategy comparison

Note: MLF = Mono, landscape, fixed tilt; HLF = HJT, landscape, fixed tilt; HLSN = HJT, landscape, seasonal tilt; HLT = HJT, landscape, two-axis tracking.

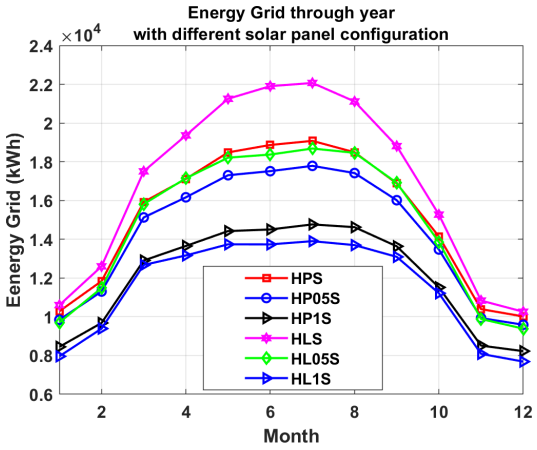


Figure 12. Monthly grid energy for portrait/landscape and spacing comparison

Note: HPS = HJT, portrait, 0 m spacing; near-shading only; HP05S = HJT, portrait, 0.5 m spacing; near-shading only; HP1S = HJT, portrait, 1.0 m spacing; near-shading only; HLS = HJT, landscape, 0 m spacing; near-shading only; HL05S = HJT, landscape, 0.5 m spacing; near-shading only; HL1S = HJT, landscape, 1.0 m spacing; near-shading only.

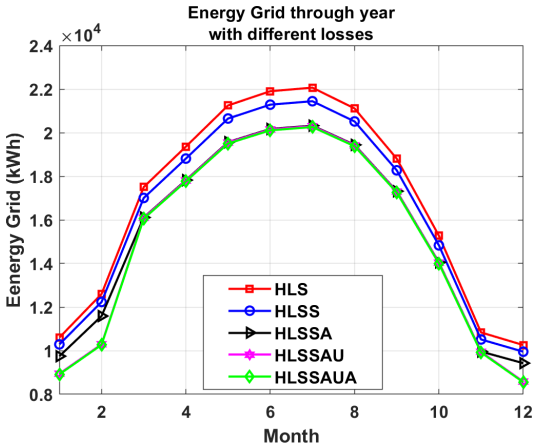


Figure 13. Monthly grid energy as realistic loss terms are added progressively

Note: HLS = HJT, landscape, 0 m spacing; near-shading only; HLSS = HJT, landscape, 0 m; shading + soiling; HLSSA = HJT, landscape, 0 m; shading + soiling + aging; HLSSAU = HJT, landscape, 0 m; shading + soiling + aging + unavailability; HLSSAUA = HJT, landscape, 0 m; shading + soiling + aging + unavailability + auxiliaries.

The influence of row distance must be considered together with installed capacity. For the landscape shading-only cases, specific yields rise from 1,544 kWh/kWp/yr at 0 m spacing (HLS) to 1,699 kWh/kWp/yr at 0.5 m spacing (HL05S) and 1,754 kWh/kWp/yr at 1.0 m spacing (HL1S), showing that additional spacing mitigates mutual shading. At the same time, annual energy falls from 201,576 to 177,922 and then to 138,335 kWh/yr because fewer modules fit on the roof. Thus, wider spacing improves output per installed kWp but reduces total electricity from the available roof area. The 0.5 m case may offer a compromise when maintenance access and shading reduction are priorities, whereas denser packing can be preferable when maximizing total annual rooftop energy is the primary objective.

Figure 13 shows the sequence of losses and why practical non-idealities should be included in the simulation. Starting from HLS (201,576 kWh/yr), adding soiling (HLSS), aging (HLSSA), unavailability (HLSSAU), and auxiliary consumption (HLSSAUA) reduces annual energy to 181,982 kWh/yr, a cumulative decrease of approximately 9.72%. This reduction is the cumulative effect of several losses rather than a single penalty. The result reinforces the importance of cleaning, reliable operation, and appropriate thermal/electrical design for rooftop PV systems in Baghdad.

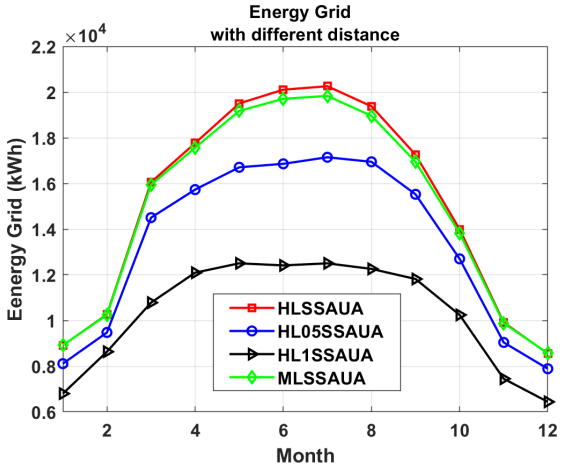


Figure 14. Monthly grid energy for different inter-row distances under fully loaded conditions

Note: HLSSAUA = HJT, landscape, 0 m; shading + soiling + aging + unavailability + auxiliaries; HL05SSAUA = HJT, landscape, 0.5 m + all added losses; HL1SSAUA = HJT, landscape, 1.0 m + all added losses; MLSSAUA = Mono, landscape, 0 m + all added losses.

3.3 Economic and Environmental Comparison

The economic results are summarized in Table 5. They show that the configuration with the highest energy output is not automatically the most attractive financial option. This distinction is essential for practical design decisions.

Table 5. Economic and environmental performance of the 17 photovoltaic (PV) variants

Variant	LCOE (USD/kWh)	Investment (USD)	Payback (yr)	NPV (USD)	IRR (%)	ROI (%)	Avoided CO_2 (tCO_2)
MLF	0.0117	45,736	3.1	239,686	31.93	524.1	3,553.8
HLF	0.0116	46,120	3.1	245,365	32.34	532.0	3,627.9
HLSN	0.0111	46,120	3.0	256,434	33.58	556.0	3,761.6
HLT	0.0179	79,720	4.9	239,649	20.32	300.6	4,043.8
HPS	0.0111	32,820	3.0	184,026	33.82	560.7	2,695.4
HP05S	0.0126	35,170	3.4	168,862	29.67	480.1	2,546.3
HP1S	0.0088	20,800	2.3	153,348	42.88	737.3	2,151.6
HLS	0.0140	46,120	3.8	192,810	26.46	418.1	2,993.4
HL05S	0.0110	31,940	2.9	180,664	34.07	565.6	2,642.1
HL1S	0.0091	20,640	2.4	145,479	41.22	704.8	2,054.3
HLSS	0.0144	46,120	3.9	185,718	25.66	402.7	2,907.8
HLSSA	0.0152	46,120	4.1	173,064	24.24	375.2	2,755.0
HLSSAU	0.0155	46,120	4.2	169,236	23.80	366.9	2,708.8
HLSSAUA	0.0155	46,120	4.2	168,710	23.74	365.8	2,702.4
HL05SSAUA	0.0122	31,940	3.3	159,412	30.65	499.1	2,385.5
HL1SSAUA	0.0102	20,640	2.7	127,779	36.82	619.1	1,840.6
MLSSAUA	0.0156	45,736	4.2	166,222	23.62	363.4	2,666.9

Note: LCOE = levelized cost of electricity; NPV = net present value; IRR = internal rate of return; ROI = return on investment.

The two-axis HJT case illustrates the distinction between maximum energy output and economic attractiveness. HLT requires the highest initial investment (79,720 USD) and delivers the maximum annual energy and avoided

operational emissions, but its simple payback extends to about 4.9 years and its discounted LCOE is approximately 0.0179 USD/kWh, higher than that of the simpler fixed and seasonal HJT cases. Conversely, some smaller spaced configurations show favorable normalized financial indicators because of their lower initial investment, although they deliver less total annual energy from the available roof. Financial indicators should therefore be interpreted together with total energy contribution, roof utilization, and constructability.

HLF remains a practical roof-scale option. It produces 244,305 kWh/yr with a discounted LCOE of approximately 0.0116 USD/kWh, a simple PBP of about 3.1 years, and an NPV of approximately 245,366 USD under the stated assumptions. HLSN produces more energy and gives a slightly better modeled financial result, but periodic tilt adjustments may be less practical under local dusty conditions. Some portrait or widely spaced cases can also appear attractive in relative financial terms because of their smaller investments, but they use less installed capacity and therefore provide lower total annual energy and avoided operational CO_2 from the roof.

Environmental performance was expressed as operational grid CO_2 avoided over the 25-year study period. Using the adopted 0.594 $kgCO_2$/kWh factor and the annual energies in Table 4, avoided emissions range from approximately 1,840.6 to 4,043.8 tCO_2. Because this operational metric is directly proportional to generated energy, HLT has the largest avoided-emission value in the present calculation. Possible life-cycle effects associated with extra tracking materials, maintenance, or component replacement are not quantified here and should therefore not be used to reverse that ranking without a separate life-cycle assessment.

3.4 Positioning Within Recent Literature and Implications for Urban Energy Transition

Table 6 positions the present study relative to representative Scopus-indexed literature. Earlier work demonstrates the value of PVsyst design analysis, urban near-shading models, HJT technology reviews, yield benchmarking, and financial evaluation, while the present study combines these elements in a single Baghdad rooftop framework.

Table 6. Comparative positioning of the present work within recent literature

Ref.	Scope/Study Type	Tools/Data	Main Technical Metrics	Main Economic Metrics	Position Relative to This Study
[15]	PV technology/tilt comparison in Tanzania	PVsyst + climatic data	Produced energy, specific production	LCOE	Broad technology and tilt comparison; not a Baghdad urban rooftop.
[6]	Rooftop PV layout optimization in Baghdad	SketchUp + PVsyst	Energy density, yield, shading	Payback / layout economics	Local 3D-shading context; the present work extends the comparison to 17 variants and a full economic/ environmental set.
[14]	Specific-yield analysis of rooftop PV systems	Measured data + irradiation data	Specific yield, energy	—	Provides measured rooftop-yield benchmarking; not Baghdad-specific.
[12]	Techno-economic evaluation of PV/wind systems in Portugal	Simulation + cost data	Energy production	LCOE, NPV, IRR, payback	Demonstrates integrated financial assessment; different climate and application.
[10, 11]	Reviews of HJT and advanced PV technologies	Review literature	Device efficiency and technology trends	—	Provide theoretical motivation for HJT selection and temperature-related advantages.
[5]	Urban-building PV and shading analysis	SketchUp + PVsyst	Shading, usable roof area, energy	—	Shows value of explicit urban geometry; different city and no full economic layer.
This work	Rooftop PV configurations in Baghdad with realistic shading	SketchUp/Skelion + PVsyst	Energy, specific yield, PR	LCOE, NPV, IRR, ROI, payback, investment, CO_2	Integrates 3D urban shading, 17 design variants, transparent financial assumptions, and an operational CO_2 comparison.

Note: PV = photovoltaic; HJT = heterojunction technology; PR = performance ratio; LCOE = levelized cost of energy; NPV = net present value; IRR = internal rate of return; ROI = return on investment; — = not reported.

The findings indicate several potential contributions of rooftop PV to Baghdad's urban energy transition.

Electricity generated close to the point of use may reduce daytime grid imports, and replacing a portion of grid electricity can reduce operational CO_2 emissions. However, this study does not include a building load profile, time-of-use grid analysis, or backup-generator dispatch model; therefore, peak-demand reduction and backup-generation displacement should be regarded as potential benefits rather than demonstrated outcomes. The results directly show that environmental performance depends on practical rooftop choices such as geometry, module spacing, shading, maintainability, and reliability.

From a practicality perspective, the fixed landscape HJT system is well suited to the case-study roof. It avoids the moving parts and control requirements of tracking, maintains a high PR, and is simpler to operate and maintain in dusty conditions. These characteristics can help the system remain close to expected performance over its service life. The environmental relevance of rooftop PV therefore depends not only on simulated yield but also on selecting a design that is maintainable, durable, and economically transparent in a real urban setting.

4 Conclusion

This work provides a techno-economic and environmental assessment of rooftop PV systems for a building in Baghdad using a SketchUp/Skelion rooftop model and PVsyst near-shading analysis. Seventeen configurations were compared under Baghdad's climatic conditions, and the simulations showed that the selected HJT module outperforms the comparable monocrystalline module because of its lower power temperature coefficient. Annual energy was maximized by the two-axis HJT system, while the seasonal HJT system achieved the highest PR. Considering energy production, cost, structural simplicity, maintenance requirements, and operational environmental benefit together, the fixed landscape HJT system remains a practical candidate for the case-study roof.

The row-spacing analysis reveals an important rooftop trade-off. Increasing clear spacing reduces mutual shading and increases specific production, but it also reduces the number of modules that can be installed and therefore lowers total rooftop energy. The preferred spacing consequently depends on whether the design objective is maximum total roof energy, maximum specific yield, or greater maintenance access. For this roof, the dense fixed landscape HJT layout provides high total energy and a strong techno-economic result, while the 0.5 m option becomes useful when access and shading reduction receive greater priority.

The avoided-emission calculation also demonstrates the operational environmental significance of rooftop PV for the Baghdad case study. Under the stated assumptions, the simulated systems avoid approximately 1,840.6–4,043.8 tCO_2 over 25 years relative to grid electricity. Future work should compare the simulations with measured operating data and evaluate sensitivity to electricity value, grid emission factor, cleaning frequency, and degradation rate. Future work should also investigate storage and load-matching options. Overall, realistic shading modelling is essential for rooftop PV design, and fixed landscape HJT systems are viable candidates for reducing operational emissions in Baghdad.

Author Contributions

Conceptualization, W.K.M. and A.H.S.; methodology, W.K.M. and A.H.S.; software, W.K.M.; validation, W.K.M. and A.H.S.; formal analysis, W.K.M.; investigation, W.K.M.; resources, A.H.S.; data curation, W.K.M.; writing—original draft preparation, W.K.M.; writing—review and editing, W.K.M. and A.H.S.; visualization, W.K.M.; supervision, A.H.S.; project administration, A.H.S. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflicts of interest.

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